Machine Learning with Graphs: Homework III

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Problem 1

(a) Laplacian Eigenvectors as a Basis

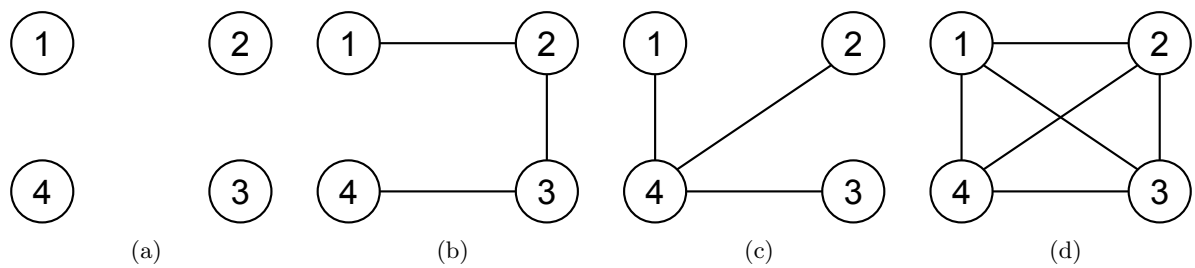

Figure 1: Problem 1(a) graphs.

To judge whether the Laplacian eigenvectors are an *effective* basis for each signal, I use the Laplacian quadratic form:

$$\mathbf{f}^\top L\mathbf{f} = \sum_{(i,j)\in E} (f_i - f_j)^2.$$

Lower values indicate smoother (lower-frequency) signals, which are represented more efficiently by the low-frequency Laplacian modes. I use two criteria: (i) smoothness (small $\mathbf{f}^\top L\mathbf{f}$) and (ii) compressibility in the Laplacian eigenbasis (how much of the signal sits in a few low-frequency coefficients). Smoothness alone does not guarantee sparse expansion.

Explicit graph matrices (node order $1, 2, 3, 4$).

$$A_a = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad L_a = D_a - A_a = 0.$$

$$A_b = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad L_b = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

$$A_c = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \quad L_c = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}.$$

$$A_d = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \quad L_d = \begin{bmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}.$$

For the three signals

$$\mathbf{f}_1 = [1, 0, 0, 0], \quad \mathbf{f}_2 = [0, 0, 0, 1], \quad \mathbf{f}_3 = [1, 1, 0, 0],$$

the smoothness values are:

Signal	Graph (a)	Graph (b)	Graph (c)	Graph (d)
$\mathbf{f}_1 = [1, 0, 0, 0]$	0	1	1	3
$\mathbf{f}_2 = [0, 0, 0, 1]$	0	1	3	3
$\mathbf{f}_3 = [1, 1, 0, 0]$	0	1	2	4

Table 1: Dirichlet energies $\mathbf{f}^\top L\mathbf{f}$.

Short algebra check (representative examples).

- Graph (b), $\mathbf{f}_1 = [1, 0, 0, 0]$:

$$\mathbf{f}_1^\top L_b \mathbf{f}_1 = [1, 0, 0, 0] \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} = 1.$$

- Graph (c), $\mathbf{f}_2 = [0, 0, 0, 1]$: only edges $(1, 4), (2, 4), (3, 4)$ cross values 0 and 1, so

$$\mathbf{f}_2^\top L_c \mathbf{f}_2 = \sum_{(i,j)\in E} (f_i - f_j)^2 = 1 + 1 + 1 = 3.$$

- Graph (d), $\mathbf{f}_3 = [1, 1, 0, 0]$: crossing edges are $(1, 3), (1, 4), (2, 3), (2, 4)$, hence

$$\mathbf{f}_3^\top L_d \mathbf{f}_3 = 4.$$

Interpretation by graph.

- **Graph (a), no edges:** $L = 0$, so all signals have zero energy and all Laplacian eigenvalues are 0. The basis is fully degenerate (no meaningful frequency ordering), so it is not informative.
- **Graph (b), path 1-2-3-4:** all three signals have energy 1 (moderately smooth), but endpoint spikes are not very compressible. For example, for $\mathbf{f}_1 = [1, 0, 0, 0]$, the squared coefficients on the four orthonormal Laplacian eigenvectors are approximately (0.25, 0.4268, 0.25, 0.0732), so all modes are active.

- **Graph (c), star centered at node 4:** $\mathbf{f}_1$ is smoothest (energy 1), $\mathbf{f}_3$ is intermediate (2), and $\mathbf{f}_2$ is roughest (3), so effectiveness depends strongly on where the nonzero entries lie relative to the hub.
- **Graph (d), complete graph K_4:** nonconstant signals are highly nonsmooth (energies 3 or 4), so these binary/local signals are not well aligned with low-frequency structure.

Projection details for the path-graph example. For P_4, an orthonormal Laplacian eigenbasis is

$$u_k(j) = \alpha_k \cos\left(\frac{(k-1)(j-\frac{1}{2})\pi}{4}\right), \quad \alpha_1 = \frac{1}{2},\ \alpha_{k>1} = \frac{1}{\sqrt{2}},$$

with $j = 1, \ldots, 4$. For $\mathbf{f}_1 = e_1 = [1, 0, 0, 0]$, coefficients are $a_k = u_k^\top e_1 = u_k(1)$, so

$$\begin{aligned}
a_1^2 &= \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}, \\
a_2^2 &= \left(\tfrac{1}{\sqrt{2}} \cos \tfrac{\pi}{8}\right)^2 = \tfrac{1}{2} \cos^2 \tfrac{\pi}{8} = \tfrac{2+\sqrt{2}}{8} \approx 0.4268, \\
a_3^2 &= \left(\tfrac{1}{\sqrt{2}} \cos \tfrac{\pi}{4}\right)^2 = \tfrac{1}{4}, \\
a_4^2 &= \left(\tfrac{1}{\sqrt{2}} \cos \tfrac{3\pi}{8}\right)^2 = \tfrac{1}{2} \cos^2 \tfrac{3\pi}{8} = \tfrac{2-\sqrt{2}}{8} \approx 0.0732.
\end{aligned}$$

Hence the coefficient energies match the values quoted above.

(b) Computing $f(\mathbf{x}) = \mathbf{x}^\top L \mathbf{x}$

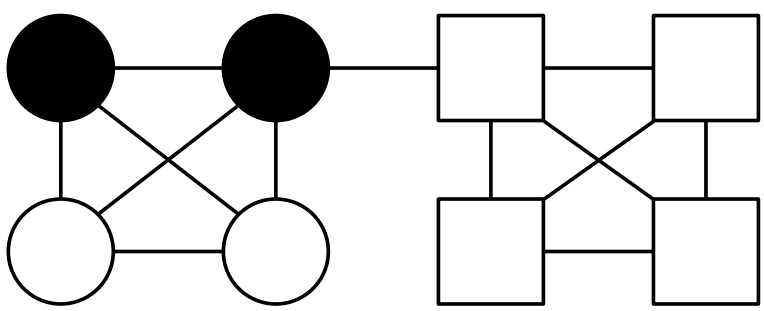

Figure 2: Problem 1(b) graph.

Using

$$\mathbf{x}^\top L \mathbf{x} = \sum_{(i,j) \in E} (x_i - x_j)^2,$$

we evaluate each case:

Constant vector. If $\mathbf{x} = c\mathbf{1}$ for any constant c, then every edge difference is zero, so

$$f(\mathbf{x}) = \mathbf{x}^\top L \mathbf{x} = 0.$$

(i) Ones on squared nodes, zeros on round nodes. All edges inside each shape group contribute 0. Only the single bridge edge between the round and square parts crosses from 0 to 1:

$$f(\mathbf{x}) = 1.$$

(ii) Ones on dark nodes, zeros on white nodes. In the left 4-node clique, there are 4 dark-white edges. The bridge edge from a dark round node to a white square node adds 1 more. Hence

$$f(\mathbf{x}) = 4 + 1 = 5.$$

Intuition. $\mathbf{x}^\top L\mathbf{x}$ measures edge-wise variation (graph total variation). For binary vectors, it equals the cut size between nodes assigned 1 and nodes assigned 0.

(c) Label Propagation on Figure 3

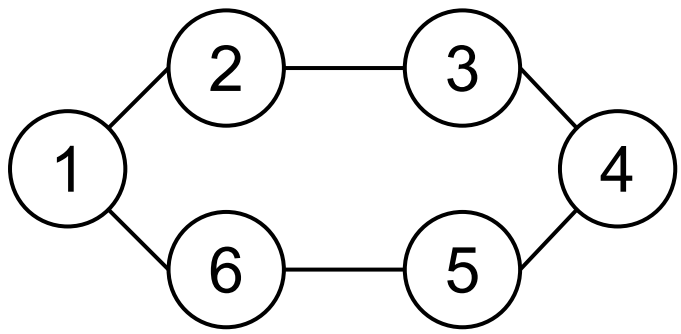

Figure 3: Problem 1(c) graph.

I use **harmonic label propagation** (real-valued labels with clamped seeds). Given labels $y_1 = 0$ and $y_4 = 1$, and unlabeled nodes $\{2, 3, 5, 6\}$, each unlabeled node is set to the average of its neighbors:

$$\begin{aligned}
y_2 &= \frac{y_1 + y_3}{2} = \frac{y_3}{2}, \\
y_3 &= \frac{y_2 + y_4}{2} = \frac{y_2 + 1}{2}, \\
y_6 &= \frac{y_1 + y_5}{2} = \frac{y_5}{2}, \\
y_5 &= \frac{y_6 + y_4}{2} = \frac{y_6 + 1}{2}.
\end{aligned}$$

Solving gives

$$y_2 = y_6 = \frac{1}{3}, \qquad y_3 = y_5 = \frac{2}{3}.$$

So the final *soft* labeling is

$$[y_1, y_2, y_3, y_4, y_5, y_6] = \left[0, \frac{1}{3}, \frac{2}{3}, 1, \frac{2}{3}, \frac{1}{3}\right].$$

Using a 0.5 threshold, the hard labels are

$$[0, 0, 1, 1, 1, 0].$$

Iterations to converge. Because the harmonic conditions form a linear system, we can solve it directly (as above), so no iterations are required for the exact answer.

If we instead run synchronous averaging (Jacobi) on the unlabeled nodes, define

$$\mathbf{x}^{(t)} = [y_2^{(t)}, y_3^{(t)}, y_5^{(t)}, y_6^{(t)}]^\top.$$

Then

$$\mathbf{x}^{(t+1)} = P\mathbf{x}^{(t)} + \mathbf{b}, \quad P = \begin{bmatrix} 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}.$$

The eigenvalues of P are $\{\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$, so $\rho(P) = \frac{1}{2} < 1$ and the iteration converges linearly. With $\mathbf{x}^{(0)} = \mathbf{0}$,

$$\|\mathbf{x}^{(t)} - \mathbf{x}^\star\|_\infty \le \frac{\rho(P)^t}{1-\rho(P)} \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty = 2^{-t}.$$

$$\text{(here } \|\mathbf{x}^{(1)} - \mathbf{x}^{(0)}\|_\infty = \|[0, \tfrac{1}{2}, \tfrac{1}{2}, 0]^\top\|_\infty = \tfrac{1}{2}\text{)}$$

Hence $t = 10$ guarantees error $\le 10^{-3}$, and $t = 20$ guarantees error $\le 10^{-6}$.

(d) Confidence-Weighted Laplacian Regularization

A standard Laplacian-regularized objective is

$$\min_{\mathbf{f}} \ \|\mathbf{f} - \mathbf{y}\|_2^2 + \mu\, \mathbf{f}^\top L \mathbf{f}.$$

To model node-specific confidence in initial labels, replace the uniform fit term with a diagonal confidence matrix $C = \mathrm{diag}(c_1, \ldots, c_n)$:

$$\min_{\mathbf{f}} \ (\mathbf{f} - \mathbf{y})^\top C (\mathbf{f} - \mathbf{y}) + \mu\, \mathbf{f}^\top L \mathbf{f}.$$

The solution is

$$\mathbf{f}^\star = (C + \mu L)^{-1} C \mathbf{y}.$$

Large c_i enforces stronger trust in node i's observed label; small c_i allows more smoothing from neighbors. In the usual semi-supervised setting, set $c_i > 0$ for labeled nodes and $c_i = 0$ for unlabeled nodes. Then unlabeled entries of $\mathbf{y}$ can be set arbitrarily (often 0), since they are multiplied by $c_i = 0$.

Toy 2-node illustration. For two nodes connected by one edge,

$$L = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} c_1 & 0 \\ 0 & c_2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}.$$

Then

$$C + \mu L = \begin{bmatrix} c_1 + \mu & -\mu \\ -\mu & c_2 + \mu \end{bmatrix},$$

and

$$\mathbf{f}^\star = \frac{1}{(c_1+\mu)(c_2+\mu) - \mu^2} \begin{bmatrix} c_2 + \mu & \mu \\ \mu & c_1 + \mu \end{bmatrix} \begin{bmatrix} c_1 y_1 \\ c_2 y_2 \end{bmatrix}.$$

This explicitly shows how larger c_i pulls $f_i^\star$ toward y_i.

In practice, c_i can be determined from:

- annotator agreement (higher agreement $\Rightarrow$ higher confidence),
- model-based uncertainty (higher predictive entropy $\Rightarrow$ lower confidence),
- source reliability or sensor quality metadata,
- local consistency checks (labels conflicting with trusted neighbors get reduced confidence).

Problem 2

We are given

$$\min_{Y} \sum_{(i,j)\in E} \|Y[i] - Y[j]\|_2^2, \qquad Y \in \mathbb{R}^{n\times D}.$$

(a) Reformulation with the Laplacian

Let $L = D - A$ be the (unnormalized) graph Laplacian. Then

$$\sum_{(i,j)\in E} \|Y[i] - Y[j]\|_2^2 = \mathrm{Tr}(Y^\top L Y).$$

So the objective is

$$\min_{Y} \ \mathrm{Tr}(Y^\top L Y).$$

(b) Computing Y from the Laplacian spectrum

The unconstrained problem has the trivial collapse solution $Y[i] = \text{constant}$ for all i. To obtain a nontrivial embedding, impose standard constraints:

$$Y^\top Y = I_D, \qquad Y^\top \mathbf{1} = 0.$$

Using a Lagrangian

$$\mathcal{L}(Y, \Lambda, \boldsymbol{\nu}) = \mathrm{Tr}(Y^\top L Y) - \mathrm{Tr}\big(\Lambda(Y^\top Y - I)\big) - 2\boldsymbol{\nu}^\top Y^\top \mathbf{1},$$

where $\Lambda = \Lambda^\top$ (WLOG for orthogonality constraints). Using matrix derivatives for symmetric L:

$$\nabla_Y \mathrm{Tr}(Y^\top L Y) = 2LY, \quad \nabla_Y \Big[-\mathrm{Tr}\big(\Lambda(Y^\top Y - I)\big)\Big] = -2Y\Lambda, \quad \nabla_Y(-2\boldsymbol{\nu}^\top Y^\top \mathbf{1}) = -2\mathbf{1}\boldsymbol{\nu}^\top.$$

Setting $\nabla_Y \mathcal{L} = 0$ gives

$$2LY - 2Y\Lambda - 2\mathbf{1}\boldsymbol{\nu}^\top = 0.$$

Premultiplying by $\mathbf{1}^\top$, using $\mathbf{1}^\top L = 0$ and $Y^\top \mathbf{1} = 0$, yields $\boldsymbol{\nu} = 0$. Therefore stationarity reduces to the eigen-condition

$$LY = Y\Lambda,$$

so each column of Y is a Laplacian eigenvector. With eigendecomposition $L = U\Lambda U^\top$ (eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n$), Courant-Fischer/Rayleigh-Ritz yields:

$$Y^\star = [u_2, u_3, \ldots, u_{D+1}],$$

where u_k is the eigenvector for λ_k.

Hence the embedding coordinates are the first D nontrivial Laplacian eigenvectors (skip the constant eigenvector u_1). If the graph is disconnected, skip the full zero-eigenspace and take the next D eigenvectors. Under these constraints, the minimum objective value is exactly

$$\mathrm{Tr}\big((Y^\star)^\top L Y^\star\big) = \sum_{k=2}^{D+1} \lambda_k$$

(or, for a graph with multiple zero eigenvalues, the sum of the chosen D smallest nonzero eigenvalues).

(c) Limitations of this embedding

- **Trivial-collapse risk without constraints:** the raw objective alone is minimized by identical node embeddings.
- **Only local smoothness is modeled:** adjacent nodes are forced close, but broader semantic structure may be missed.
- **Sensitivity to graph quality:** noisy, missing, or spurious edges directly distort the embedding.
- **Disconnected graphs:** multiple zero eigenvalues create ambiguity and can fragment embeddings.
- **Scalability:** eigendecomposition is expensive for large graphs.
- **No supervision/attributes in the basic form:** node features and labels are ignored unless explicitly added.
- **Weak multiscale expressiveness:** a single global quadratic objective may miss hierarchical/community structure across scales.